# HARSHAL RAAJ PATNAIK

**Senior Software Engineer & Platform Architect | Cloud Infrastructure, Distributed Systems & Product Strategy (Tech × MBA)**

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## EDUCATION

**Indian Institute of Management Kozhikode (IIMK)** **2025 – Present**
*Executive Post Graduate Programme (EPGP) in General Management – Product Strategy, Finance & Operations* *Kochi, India*

**SRM Institute of Science and Technology** **2017 – 2021**
*Bachelor of Technology in Computer Science and Engineering (CGPA: 8.76 / 87.6%)* *Chennai, India*

## PROFESSIONAL EXPERIENCE

**Novamesh Ltd. (Wholly Owned Subsidiary of Tata Communications)** **Apr 2024 – Present**
*Senior Software Engineer (Cloud Platforms & Distributed Infrastructure)* *Chennai, India*

- **Scaled VM Dynamic Autoscaling:** *Scaled* dynamic VM horizontal and vertical autoscaling policies across **KVM, VMware ESXi, and OLVM** hypervisors based on real-time CPU/memory thresholds, preventing resource starvation during traffic bursts.
- **Built Automated Order-to-Deployment Pipeline:** *Built* end-to-end zero-touch order fulfillment pipeline converting customer service orders into automated, self-healing cloud deployments; integrated **Flowable BPMN sagas** and **RabbitMQ** to orchestrate VM provisioning, storage provisioning, network routing, and IP assignment with automated rollbacks, slashing customer provisioning lead time by **85% (from 4 hours to under 35 minutes)**.
- **Designed Capacity & Infrastructure Dashboard:** *Designed* and engineered full-stack multi-tenant Capacity Planning & Inventory Dashboard from design to distributed backend services, managing real-time lifecycle records across **Compute** (KVM, VMware ESXi, OLVM hypervisors), **Storage** (NetApp and HPE storage arrays, Caringo Swarm), and **Network** (public IP pool reports, switch port mappings, VDOM reports, bandwidth utilization telemetry) with interactive graphical heatmaps and trend analytics.
- **Engineered High-Throughput Batch Processing:** *Engineered* batch processing optimizations slashing cloud scheduler execution latency by **92.6% (from ~18 minutes down to 80 seconds)** across distributed Cassandra clusters using strict 50-item chunking, multi-level Redis caching, and HikariCP connection pool tuning.
- **Led Kong API Gateway & Fortinet Firewall Automation:** *Led* enterprise edge routing and security automation across **Kong API Gateway** and **Fortinet FortiGate Firewalls**; engineered dynamic route discovery, wildcard forwarding, and IPsec VPNs via REST APIs while implementing an in-process sliding-window rate limiter that **eliminated 100% of HTTP 429 throttling errors**.

**Tata Communications Ltd.** **Jul 2021 – Mar 2024**
*Software Development Engineer (Cloud Infrastructure, Storage & Full-Stack)* *Chennai, India*

- **Built Cloud Object Storage & DNS Automation:** *Built* full-stack onboarding workflow for **Caringo Swarm S3-compatible cloud object storage** and automated cPanel DNS A-record associations using Java and Angular, enabling **100% self-service storage provisioning and automated bucket metering**.
- **Scaled Frontend Modernization:** *Scaled* legacy AngularJS migration to **Angular 17** with reactive component architectures, state predictability, and modular testability, achieving a **35% improvement in client-side rendering speed**.
- **Automated CI/CD & Multi-Cloud Deployment:** *Automated* continuous integration, regression testing, and release packaging pipelines with **Jenkins CI/CD**, multi-module Maven builds, and JUnit quality gates, significantly cutting manual verification overhead and deployment friction.
- **Analyzed Network Telemetry & Monitoring:** *Analyzed* and engineered switch-port utilization and network capacity telemetry engine integrating Zabbix and Nimsoft APIs, computing cluster-wide compute and datastore allocation to prevent capacity exhaustion; earned **3× Extra Mile Commitment Awards**.
- **Architected Multi-Vendor Load Balancer Orchestration:** *Architected* enterprise load balancer provisioning and health probe monitoring across **HAProxy, F5 BIG-IP, and VMware Avi** with automated VIP provisioning, traffic steering, and tenant-scoped SSL/TLS offloading.

## STRATEGY & TECHNICAL PROJECTS

**SthiraX: Distributed $CO_2$-Mineralization & IoT MRV Platform | *IIM Kozhikode Capstone (BPP)*** **GitHub ↗**

- *Architected* asset-light retrofit platform converting regional concrete plants into certified low-carbon facilities to fulfill India's **April 2026 C&D Waste Rules (mandating 5% to 25% recycled material by 2030 across 20,000+ sqm projects)**; built batch-level **IoT MRV (Measurement, Reporting & Verification) software** generating cryptographically verifiable Digital Product Passports for developer EPR compliance, supported by SaaS and royalty unit economics.

**MarginOS: Multi-Tenant Cloud FinOps & Unit Economics Platform | *Cloud Economics & FinOps*** **GitHub ↗**

- *Designed* distributed FinOps intelligence engine analyzing multi-cloud hypervisor and storage telemetry to calculate real-time gross margins, compute unit economics (cost-per-VM/hour, storage tiers), and surface automated right-sizing recommendations that eliminate up to 28% in infrastructure spend leakages.

**Workflow Reliability Lab: Distributed Sagas & Chaos Engineering | *Distributed Systems Platform*** **GitHub ↗**

- *Built* developer platform for testing distributed workflow reliability; built synthetic fault injection simulators (network partitions, timeout anomalies) and invariant checkers verifying ACID saga compensations (**Java 25, Spring Boot 4, Angular, Docker**).

**QueryLens AI: Sandboxed Enterprise Text-to-SQL & Semantic RAG | *AI & Data Systems*** **GitHub ↗**

- *Architected* modular C4 3-tier Text-to-SQL engine (**Angular 17 → Spring Boot 3 → FastAPI with pgvector RAG**) with schema semantic search, AST validation, and read-only sandbox roles preventing SQL injection and unauthorized data exfiltration.

**CloudScale: Predictive Workload Autoscaler & Telemetry Engine | *Autonomous Cloud Infrastructure*** **GitHub ↗**

- *Developed* proactive Kubernetes/VM capacity autoscaling engine integrating Prometheus, Grafana, and time-series ML forecasting models predicting compute resource spikes 15 minutes in advance, mitigating cold-start latencies and reducing over-provisioned idle capacity.

## TECHNICAL & LEADERSHIP COMPETENCIES

**Languages & Core:** Java (8/11/17/25 LTS), Python (NumPy, Pandas, Scikit-learn, Matplotlib), TypeScript, JavaScript, SQL, HTML5/CSS3, Bash, Spring Boot 3/4, FastAPI, Angular 17/18, RxJS

**Distributed Systems & Cloud:** Google Cloud Platform (GCP / Compute Engine), Microservices, Flowable BPMN, RabbitMQ (AMQP), Event-Driven Architecture, High Availability (HA), Fault Tolerance, KVM, VMware ESXi, OLVM, Docker, Kubernetes

**Databases, Caching & Storage:** Apache Cassandra (NoSQL), Redis (In-Memory Cache), PostgreSQL (pgvector), MySQL, Caringo Swarm (S3 Object Storage), NetApp & HPE Storage Arrays, Hibernate/JPA, HikariCP Tuning

**Networking, Security & Gateways:** Kong API Gateway, Fortinet FortiGate Firewalls, VDOMs, IPsec VPN, DNS Automation (cPanel/BIND), Token Bucket Rate Limiting, Reverse Proxies, HAProxy, F5 BIG-IP, VMware Avi

**DevOps, SRE & Observability:** Jenkins CI/CD, CI/CD Pipelines, Prometheus, Grafana, OpenTelemetry, Distributed Tracing, Zabbix APIs, Nimsoft, JUnit 5, Mockito, SonarQube, Linux/Unix Administration

**Product Management & MBA Coursework:** Product Strategy, Financial Accounting, Operations Management, Strategy Management, Risk Analysis, Product Analytics, Digital Marketing, Brand Strategy, Negotiation & Conflict Management, FinOps, Jira (Agile/Scrum)

## LEADERSHIP, CERTIFICATIONS & HONORS

- **Tech × MBA Strategic Product Leadership:** Bridge the critical organizational gap between platform economics and deep systems engineering; partner with executive product leaders and enterprise clients to translate business roadmaps and unit-cost targets into resilient distributed microservices, author technical RFCs, and govern SLA/SLO availability baselines.
- **Agile Engineering Delivery & Mentorship:** Stepped up as **Acting Scrum Master** across engineering pods; orchestrated sprint ceremonies, backlog prioritization, and release velocity in **Jira** (100% on-time milestone delivery), while providing architectural reviews and technical mentorship for junior engineers.
- **3× Extra Mile Commitment Award** – Tata Communications Ltd. for critical cloud platform delivery, stability, and zero-defect deployments.
- **Flipkart Grid 2.0 Engineering Challenge** – Team Lead; achieved **Top 150 Nationwide Rank** out of 20,000+ teams (Computer Vision & OCR Track).
- **Certifications: Architecting with Google Compute Engine** (Google Cloud / Coursera) & **Full Stack Web Development** (Coding Ninjas).